Tentative Course Outline

Course Name	Machine Learning
Course Code	CS-XXX (as per program scheme)
Credit Hours	3 (Theory Credit Hour = 2, Lab Credit Hour =1)
Contact Hours	2 hrs Theory + 3 hrs Lab per week
Pre-requisites	Artificial Intelligence
Program	BS-V (CS, SE & AI) – Sections (B, C, D, E, F, G & H)
Duration	16 Teaching Weeks + 2 Exam Weeks
Course Teachers	Dr Muhammad Ismail Mangrio, Dr Zakriya, Dr Adil Khan & Dr Saif Hassan

Course Objectives

This course aims to introduce students to the fundamentals of machine learning and its role as one of the fastest-growing areas of computer science. It seeks to build a solid conceptual foundation by presenting the basic principles, terminology, and theories that underpin the field, before moving on to a broad range of supervised, unsupervised, and reinforcement learning algorithms, examined critically in terms of their respective strengths and weaknesses. Throughout the course, students will apply these algorithms to problems of moderate complexity using real datasets, developing the practical, hands-on skills needed to select, implement, and evaluate an appropriate machine learning technique for a given problem culminating in a course project that demonstrates their ability to design and build a complete, reasonably sized machine learning solution.

Week-by-Week Teaching Plan

Week	Lecture Topic(s)	CLO(s)	Lab Session
1	Introduction to Machine Learning: definitions, types of learning (supervised, unsupervised, reinforcement, semi-supervised), real-world applications, ML pipeline overview. Python/NumPy environment setup.	CLO-1	Lab 1 — Python Refresher & NumPy Fundamentals, Data Handling with Pandas, Data Visualization with Matplotlib
~~2~~	Supervised Learning I: Decision Trees, entropy, information gain, ID3 algorithm; overfitting and noisy data.	CLO-2	Lab 2 — Data Preprocessing & Scikit-learn Basics
~~3~~	Decision Trees continued: pruning strategies, handling continuous attributes, CART.	CLO-2	Lab 3 — Decision Tree Classifier
4	Naive Bayes Classifier: Bayes theorem, conditional independence assumption, text	CLO-2	Lab 4 — Naive Bayes Classifier

Week	Lecture Topic(s)	CLO(s)	Lab Session
	classification example.		
5	Linear and Logistic Regression: cost functions, gradient descent, regularization (L1/L2). Bias vs Variance.	CLO-2	Lab 5 — Linear Regression and Logistic Regression
6	Support Vector Machines: maximum margin classifiers, kernels (linear, polynomial, RBF), soft margin.	CLO-2	Lab 6 — Support Vector Machines (Kernels)
7	Measuring Classifier Accuracy: confusion matrix, precision/recall/F1, ROC-AUC, cross-validation.	CLO-2	Lab 7 — Model Evaluation & Classifier Accuracy Metrics
8	Midterm Examination Week — Review Session covering Weeks 1-7.		Mid Exam
9	Unsupervised Learning I: Hierarchical Agglomerative Clustering, k-means partitional clustering.	CLO-3	Mid Exam Lab
10	k-means partitional clustering.	CLO-3	Lab 10 — Clustering: K-Means & Hierarchical Agglomerative Clustering K-Nearest Neighbors & Self-Organizing Maps
11	Artificial Neural Networks I: the perceptron, activation functions, gradient descent basics.	CLO-2	Lab 11 — Artificial Neural Networks (Perceptron)
12	Artificial Neural Networks II: multi-layer perceptron, backpropagation, overfitting in neural nets.	CLO-2	Lab 12 — Multi-layer Perceptron
13	Ensemble Learning: bagging, boosting, committees of hypotheses. Random Forest, XGBoost		Lab 13 — Ensemble Learning: Bagging & Boosting
14	Reinforcement Learning: Hidden Markov Models, Markov Decision Processes, exploration vs. exploitation trade-off, Monte Carlo inference.	CLO-4	Lab 14 — Reinforcement Learning Basics: MDPs & Q-Learning
15	Reinforcement Learning continued: Q-Learning basics. Course wrap-up, final project presentations.	CLO-4, CLO-5	Lab 15 — Project Presentations
16	Lab Exam		

Textbooks and Reference Materials

1. Mitchell, T. M. Machine Learning. McGraw Hill, 1997.
2. Murphy, K. P. Machine Learning: A Probabilistic Perspective. MIT Press, 2012.
3. Geron, A. Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd ed.). O'Reilly, 2022. (Recommended for lab sessions)
4. James, G., Witten, D., Hastie, T., Tibshirani, R. An Introduction to Statistical Learning. Springer, 2021.

Assessment Plan

Assessment Component	Weight	Notes
Quizzes (announced/unannounced)	**5%**	Weekly, aligned to preceding lecture topic
Assignments	**5%**	2-3 assignments covering theory + short implementation tasks + 2-3 quiz
Mid Term Examination	**30%**	Week 8 — covers Weeks 1-7
Course Project (CLO-5)	**10%**	Proposal (Week 9), Progress check (Week 13), Final demo/report (Week 16)
Final Term Examination	**50%**	Comprehensive, covers full course

Course Learning Outcomes (CLOs)

CLO No.	Course Learning Outcome	Bloom Taxonomy
CLO-1	Describe basic machine learning concepts, theories and applications.	C1 (Knowledge)
CLO-2	Apply supervised learning techniques to solve classification and regression problems of moderate complexity.	C3 (Apply)
CLO-3	Apply unsupervised learning techniques to solve clustering problems of moderate complexity.	C3 (Apply)
CLO-4	Apply reinforcement learning algorithms to environments with complex dynamics.	C3 (Apply)
CLO-5	Develop a reasonable size project using a suitable machine learning technique.	C6 (Create)

CLO – GA Mapping

The matrix below maps each Course Learning Outcome (CLO) to the Graduate Attribute(s) (GAs) it primarily contributes to, using a binary scheme (1 = mapped, 0 = not mapped).

CLO ID	Graduate Attributes (GAs)									
	GA1	GA2	GA3	GA4	GA5	GA6	GA7	GA8	GA9	GA10
CLO-1	**1**	0	0	0	0	0	0	0	0	0
CLO-2	0	**1**	0	0	**0**	0	0	0	0	0
CLO-3	0	0	**1**	0	**0**	0	0	0	0	0
CLO-4	0	0	0	**1**	**0**	0	0	0	0	0
CLO-5	0	0	0	**0**	**1**	**0**	**0**	0	**0**	**0**

Program Learning Outcomes / Graduate Attributes (GAs)

This degree program prepares students to attain educational objectives by ensuring that students demonstrate achievement of the following outcomes (derived from Graduate Attributes defined by Seoul Accord www.seoulaccord.org).

S. No.	Graduate Attributes (GAs)	Computing Professional Graduate Outcomes
1	**Academic Education**	To prepare graduates as CS professionals.
2	**Knowledge for Solving Computing Problems**	Apply knowledge of CS, knowledge of a CS specialization, and mathematics, science, and domain knowledge appropriate for the CS specialization to the abstraction and conceptualization of CS models from defined problems and requirements.
3	**Problem Analysis**	Identify, formulate, research literature, and solve complex CS-related problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines.
4	**Design / Development of Solutions**	Design and evaluate solutions for complex CS problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
5	**Modern Tool Usage**	Create, select, adapt, and apply appropriate techniques, resources, and modern computing tools to complex CS activities, with an understanding of the limitations.
6	**Individual and Team Work**	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings.
7	**Communication**	Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend

S. No.	Graduate Attributes (GAs)	Computing Professional Graduate Outcomes
		and write effective reports, design documentation, make effective presentations, and give and understand clear instructions.
8	**CS Professionalism and Society**	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional CS practice.
9	**Ethics**	Understand and commit to professional ethics, responsibilities, and norms of professional CS practice.
10	**Life-long Learning**	Recognize the need, and have the ability, to engage in independent learning for continual development as a CS professional.

Approvals

Prepared By	Dr Muhammad Ismail Mangrio, Dr Zakriya, Dr Adil & & Dr Saif Hassan
Approved By	Not Specified
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